TRANSFORMER SUMMARIZER — COMPLETE EVALUATION PIPELINE

One article → 3 models → full step-by-step transformations → 4 evaluation modules → metrics dashboard

STAGE 1

STAGE 1 — INPUT ARTICLE (CNN/DailyMail dataset)

ARTICLE (same example used in ALL stages of this poster):

"Scientists discovered a massive black hole located near Earth,
raising new questions about galactic formation. The team, led by Dr. Sarah Chen at MIT,
used the James Webb Space Telescope. Mass: 10 billion x our Sun. Formed in early universe."

CORPUS	SIZE	AVG LEN	VOCAB	SPLIT
CNN/DailyMail	**280,327 articles**	**~780 tokens**	**~50k BPE**	**train/val/test**

REFERENCE SUMMARY

Gold standard (human-written):

"Scientists found a huge black
hole close to Earth. MIT team
led by Dr Chen used Webb."

Used ONLY for intrinsic eval:

ROUGE + NER scoring

(not needed for Sentiment/Topic)

reference ->ROUGE, NER

extractive

abstractive (sequential)

abstractive (parallel)

STAGE 2

(1) TF-IDF EXTRACTIVE BASELINE

APPROACH: Extractive — copies full sentences verbatim

No neural network. Scores each sentence by keyword importance (TF-IDF) then copies top-K as the summary.

Key formulas:

```
TF(w,s) = count(w in s) / |s|          (term freq)
IDF(w) = log( N / df(w) )               (inv doc freq)
TF-IDF(w,s) = TF(w,s) x IDF(w)
score(sentence) = SUM of TF-IDF over content words
```

STEP-BY-STEP TRANSFORMATION TRACE

INPUT: Article Sentence S1

```
"Scientists discovered a massive black hole near Earth."
```

STEP 1 — TOKENIZE + REMOVE STOP WORDS

full: scientists discovered a massive black hole near earth

stop words removed: "a" (the, is, was, of, to... also removed)

```
content tokens: scientists discovered massive black hole near earth
|S1| = 8 content words
```

STEP 2 — TERM FREQUENCY TF = count / |S1|

Every word appears once → TF = 1/8 = 0.125 for all words

```
TF("scientists") = 1/8 = 0.125
TF("hole")       = 1/8 = 0.125    (same for all 8 words)
```

STEP 3 — INVERSE DOCUMENT FREQUENCY IDF = log(N/df)

N = 280,327 total articles in corpus

```
Word        df (articles)   N/df      IDF = log(N/df)
scientists      1,200        233.6   = 5.45   <- rare, informative
discovered      8,500         33.0   = 3.50
massive         4,200         66.7   = 4.20
black          12,000         23.4   = 3.15   <- common, low IDF
hole              800        350.4   = 5.86   <- very rare, high!
near           15,000         18.7   = 2.93   <- common word
earth           3,000         93.4   = 4.54
```

STEP 4 — TF-IDF PRODUCT = TF x IDF

```
scientists = 0.125 x 5.45 = 0.681
discovered = 0.125 x 3.50 = 0.438
massive    = 0.125 x 4.20 = 0.525
black      = 0.125 x 3.15 = 0.394
hole       = 0.125 x 5.86 = 0.733  <- highest score in sentence
near       = 0.125 x 2.93 = 0.366
earth      = 0.125 x 4.54 = 0.568
```

STEP 5 — SENTENCE SCORE = SUM of TF-IDF values

```
score(S1) = 0.681+0.438+0.525+0.394+0.733+0.366+0.568 = 4.175
score(S1) = 4.175  -> RANK 1  SELECTED
score(S2) = 3.680  -> RANK 2  SELECTED   [Dr Chen, MIT, Webb]
```

score(S3) = 2.140 -> RANK 3 not selected

STEP 6 — OUTPUT: Copy S1 and S2 verbatim from article

```
OUTPUT: "Scientists discovered a massive black hole..."
```

PROBLEM: "discovered" kept as-is. Reference uses "found".

ROUGE: zero overlap on this word -> low recall!

`ROUGE-1: 17.0%   ROUGE-2: 6.1%   ROUGE-L: 9.2%`

(2) BiLSTM + BAHDANAU ATTENTION

APPROACH: Abstractive — encoder-decoder + additive attention

Bidirectional LSTM encoder captures left + right context.

Bahdanau single-head attention focuses decoder per step.

Key equations:

```
h_i = [LSTM_fwd(x_i,h_{i-1}) | LSTM_bwd(x_i,h_{i+1})]
e(t,i) = v * tanh(W_s*s_t + W_h*h_i)     (alignment)
a(t,i) = softmax(e(t,:))                  (attention wt)
c_t = SUM_i a(t,i)*h_i                    (context vec)
```

STEP-BY-STEP TRANSFORMATION TRACE

INPUT: tokenize article into word embeddings (dim=256)

```
tokens: [<SOS>, scientists, discovered, massive,
        black, hole, near, earth, <EOS>]   n=9
embed("scientists") = [0.82, 0.31,-0.14, ...] 256-dim
```

ENCODER — BIDIRECTIONAL LSTM (hidden_size=256)

POSITION t=1 : "scientists"

```
Forward : h1_f = LSTM_f(x1, h0=0) = [0.82,-0.21,0.54,...] 256d
Backward: h1_b = LSTM_b(x1, h9  ) = [-0.31,0.73,0.19,...] 256d
Combined: h1   = concat(h1_f, h1_b)              512-dim
```

POSITION t=2 : "discovered"

```
h2_f = LSTM_f(x2, h1_f)   <-- USES h1_f (sequential!)
     = [0.71,-0.34,0.62,...] 256d
```

Cannot compute h2_f until h1_f is done -> no parallelism

```
h2 = concat(h2_f, h2_b)                          512-dim
```

DECODER — STEP 1 : generate first output token

Initial decoder state: s0 = zeros(256)

BAHDANAU ATTENTION SCORES e(t=1, i):

```
e(1,1) = v*tanh(W_s*s0 + W_h*h1) = 0.72  "scientists"
e(1,2) = v*tanh(W_s*s0 + W_h*h2) = 0.68  "discovered"
e(1,3) = v*tanh(W_s*s0 + W_h*h3) = 0.31  "massive"
e(1,4) = v*tanh(W_s*s0 + W_h*h4) = 0.28  "black"
alpha = softmax([0.72, 0.68, 0.31, 0.28,...])
alpha = [0.38, 0.35, 0.12, 0.08, ...]  <- "scientists" + "discovered" win
context c1 = 0.38*h1 + 0.35*h2 + 0.12*h3 + ... (512d)
s1 = LSTM(concat(embed(<SOS>), c1), s0)
P(w|s1,c1): researchers=0.34  scientists=0.21  team=0.18
-> Emit: "researchers"  (paraphrase of "scientists")
```

DECODER — STEP 2 : second output token

State s1 -> recompute attention alpha2:

```
e(2,2) = score(s1, h2) = 0.81  "discovered" focus!
alpha2 = [0.15, 0.43, 0.18, 0.14, ...]
c2 = weighted sum toward h2 = "discovered"
P(w|s2,c2): found=0.41  discovered=0.12
-> Emit: "found"   (paraphrase of "discovered"!)
```

SEQUENTIAL LIMITATION

Step t+1 needs s_t -> cannot parallelise decoder

Long sequences: early context may fade (vanishing gradient)

```
OUTPUT: "researchers found large black hole near earth..."
```

`ROUGE-1: ~20%     ROUGE-2: ~8%     ROUGE-L: ~14%   (approx)`

(3) TRANSFORMER Seq2Seq

APPROACH: Abstractive — pure attention, no recurrence

All positions processed in PARALLEL via multi-head self-attention. 4 heads specialize in different relation types.

Key equations:

```
Q=X*W_Q, K=X*W_K, V=X*W_V        (d_model=512, d_k=128)
Attn(Q,K,V) = softmax(Q*Kt/sqrt(d_k)) * V
MultiHead  = Concat(head1,...,head4) * W_O
PE(pos,2i) = sin(pos/10000^(2i/512))    (positional enc)
```

STEP-BY-STEP TRANSFORMATION TRACE

INPUT: BPE tokenization + positional encoding

```
tokens: [scientists, discov-ered, a, massive, black, hole, near, earth]
        pos:    1          2      3     4       5     6    7      8
PE(1,0) = sin(1) = 0.841   PE(1,1) = cos(1) = 0.540
X[1] = embed("scientists") + [0.841, 0.540, ...]  512-dim
```

COMPUTE Q, K, V -- ALL 8 POSITIONS AT ONCE

```
Q = X * W_Q    [8x512] * [512x512] = [8x512]
K = X * W_K    [8x512] * [512x512] = [8x512]
V = X * W_V    [8x512] * [512x512] = [8x512]
Split into 4 heads: Q1..Q4, K1..K4, V1..V4  each [8x128]
```

HEAD 1 — SYNTACTIC (subject-verb, collocations)

```
scores1 = Q1 * K1_T / sqrt(128)     [8x8 matrix]
scores1["scientists","discov-ered"] = 4.2  <- subject-verb bond!
scores1["black","hole"]             = 4.5  <- collocation bond!
alpha1 = softmax(scores1)   head1 = alpha1 * V1  [8x128]
```

HEAD 2 — SEMANTIC / SYNONYM KNOWLEDGE

```
scores2["discov-ered","found"]  = 3.8  <- synonym pairing!
scores2["massive","huge"]       = 3.6  <- synonym pairing!
scores2["near","close"]         = 3.4  <- synonym pairing!
Head 2 enables Transformer to GENERATE synonyms
```

HEAD 3 — CO-REFERENCE RESOLUTION

```
scores3["MIT","team"]           = 3.9  <- same entity
scores3["Chen","she"]           = 3.7  <- pronoun link
```

HEAD 4 — LOCAL N-GRAM CONTEXT

```
scores4["black","hole"]         = 4.5  <- fixed bigram
scores4["James","Webb"]         = 4.3  <- named entity
```

```
MultiHead = Concat(h1,h2,h3,h4) * W_O  [8x512]
ALL 8 positions done simultaneously -> no decay!
```

DECODER — CROSS-ATTENTION generates output tokens

Pos Encoder focus Generates

```
1:  scientists (0.82)      ->  "Scientists"
2:  discov-ered (0.79)     ->  "found"     <- HEAD 2 synonym!
3:  a (0.91)               ->  "a"
4:  massive (0.77)         ->  "huge"      <- HEAD 2 synonym!
5:  black (0.88)           ->  "black"
6:  hole (0.92)            ->  "hole"
7:  near (0.81)            ->  "close"     <- HEAD 2 synonym!
OUTPUT: "Scientists found a huge black hole close to Earth."
3 paraphrases via Head 2 -> near-perfect match to reference!
```

`ROUGE-1: 24.5%   ROUGE-2: 9.8%   ROUGE-L: 19.9%  * BEST`

STAGE 3

TF-IDF EXTRACTIVE SUMMARY

```
"Scientists discovered a massive black hole located near Earth."
"The team, led by Dr. Sarah Chen at MIT, used James Webb..."
```

-> 2 top-scoring sentences copied VERBATIM

-> "discovered" unchanged (ref uses "found") -> ROUGE miss

-> Preserves all named entities (Sarah Chen, MIT, Webb)

BiLSTM + BAHDANAU SUMMARY

```
"researchers found large black hole near earth."
"dr chen at mit led study using webb telescope."
```

-> "researchers" paraphrase of "scientists"

-> "found" paraphrase of "discovered" -> ROUGE match!

-> Lowercase output -> NER misses "earth", "dr chen"

TRANSFORMER SUMMARY

```
"Scientists found a huge black hole close to Earth."
"MIT team led by Dr Chen used James Webb telescope."
```

-> discovered->found, massive->huge, near->close (HEAD 2!)

-> Matches reference almost exactly -> high ROUGE

-> Lowercase BPE: "earth" -> NER casing issue

Each summary enters all 4 evaluation modules independently

STAGE 4

(1) N-GRAM OVERLAP [INTRINSIC — ROUGE]

NEEDS REFERENCE

WHAT IT MEASURES

Lexical overlap between generated summary and human reference.

No semantic understanding — purely word-level matching.

ROUGE-1 (unigram F1)

```
Precision P = |{words in summary} intersect {words in ref}| / |summary|
Recall    R = |{words in summary} intersect {words in ref}| / |ref|
F1        = 2*P*R / (P+R)   <- harmonic mean
```

WORKED EXAMPLE (Transformer summary vs reference):

```
Summary : "Scientists found a huge black hole close to Earth"   n=9
Reference: "Scientists found a huge black hole close to Earth"  n=9
Overlap (unigrams): scientists found a huge black hole close earth
|S intersect R| = 8,   P = 8/9 = 0.89,   R = 8/9 = 0.89
ROUGE-1 F1 = 2*0.89*0.89/1.78 = 0.89 = 89%  (mini example)
```

Real corpus avg across test set -> Transformer: 24.5%

ROUGE-2 (bigram F1)

Uses pairs of consecutive words. One synonym difference breaks ALL bigrams containing that word!

```
TF-IDF summary: "scientists discovered a massive..."
Reference:      "scientists found a huge..."
```

(scientists,discovered) != (scientists,found) <- bigram miss!

(discovered,a) != (found,a) <- bigram miss!

```
That is why ROUGE-2 (6.1%) << ROUGE-1 (17.0%) for TF-IDF
```

ROUGE-L (Longest Common Subsequence)

```
LCS = longest sequence of matching words (not necessarily contiguous)
Precision_L = LCS/|summary|,  Recall_L = LCS/|ref|
```

Rewards preserving word ORDER, not just individual words.

```
TF-IDF copies article sentence order != reference order
```

-> LCS with reference is tiny -> ROUGE-L = 9.2%

```
Transformer output mirrors reference compressed structure
-> LCS much longer -> ROUGE-L = 19.9%  (+117% vs TF-IDF!)
```

WHY HARMONIC MEAN? (F1 vs arithmetic avg)

```
Arithmetic avg(1.0, 0.0) = 0.50   <- cheatable by copying all
Harmonic    2*1.0*0.0/(1.0+0.0) = 0.0  <- catches it!
```

Harmonic mean PENALIZES extreme imbalance between P and R.

RESULTS SUMMARY

System	ROUGE-1	ROUGE-2	ROUGE-L	
TF-IDF	17.0%	6.1%	9.2%	
BiLSTM	~20.0%	~8.0%	~14.0%	
Transformer	24.5%	9.8%	19.9%	BEST

Transformer DOUBLES TF-IDF on ROUGE-L: +117%

(2) NER ENTITY PRESERVATION [INTRINSIC]

NEEDS ARTICLE

WHAT IT MEASURES

Are key named entities from the article present in summary?

Checks factual completeness — did we lose important names/places?

PIPELINE

Step 1: Run spaCy NER on article -> Article Entities (A)

Step 2: Run spaCy NER on summary -> Summary Entities (S)

Step 3: Compute precision P = |S intersect A|/|S|
and recall R = |S intersect A|/|A|

ENTITY TYPES TRACKED

`PERSON  ORG  GPE  DATE  FACILITY  NORP  PRODUCT  EVENT`

WORKED EXAMPLE (article entities):

```
Sarah Chen    [PERSON]     <- researcher name
MIT           [ORG]        <- institution
Earth         [GPE]        <- geo-political entity
James Webb    [FACILITY]   <- telescope name
Article entity set A = {Sarah Chen, MIT, Earth, James Webb}  |A|=4
```

NER SCORES — TF-IDF extractive:

```
TF-IDF copies text verbatim -> all casing preserved
Summary entities S = {Sarah Chen, MIT, Earth, James Webb}
|S intersect A| = 4,   P = 4/4 = 100%,   R = 4/4 = 100%
TF-IDF NER: Precision=100%, Recall=100%  <- perfect
```

NER SCORES — Transformer:

```
Transformer output: "Scientists found a huge black hole..."
```

BPE vocabulary built on lowercased text!

Output: "earth" not "Earth", "dr chen" not "Dr Chen"

spaCy rule-based NER: "earth" -> not recognized as GPE

spaCy rule-based NER: "dr chen" -> may miss PERSON

```
Summary entities S = {} or S = {MIT}  (partial)
```

NER Recall = 0% (casing artifact, NOT a model error!)

FIX APPROACHES

```
1. Capitalise first char of each output token (post-process)
2. Use neural NER (case-insensitive, e.g. spaCy lg model)
3. Apply truecasing model before NER
```

KEY INSIGHT

Extractive = perfect NER (copies exact text).

Abstractive = NER depends on output casing/vocabulary.

NER recall alone does not measure summary quality.

(3) SENTIMENT CONSISTENCY [EXTRINSIC]

NO REFERENCE NEEDED

WHAT IT MEASURES

Does the summary preserve the emotional tone of the article?

Extrinsic: test on downstream sentiment classification task.

CLASSIFIER: LOGISTIC REGRESSION

```
Input: TF-IDF vector over ~5000 vocab words (sparse)
Classes: POSITIVE | NEUTRAL | NEGATIVE
Training: labeled news article dataset
```

STEP 1 — TF-IDF VECTORIZATION

1. Tokenize text, remove stop words
2. Build vocabulary: top 5000 words by corpus TF-IDF
3. Each document -> sparse vector [dim=5000]

```
   x["scientists"] = 0.82,  x["discovered"] = 0.74
   x["black"]      = 0.61,  x["hole"]       = 0.71
```

STEP 2 — LOGISTIC REGRESSION + SOFTMAX

```
For each class k:  z_k = W_k . x + b_k
z_pos  = w_pos  . x = 1.204
z_neu  = w_neu  . x = 0.910
z_neg  = w_neg  . x = 0.634
P(k|x) = exp(z_k) / [exp(z_pos)+exp(z_neu)+exp(z_neg)]
exp(1.204)=3.334,  exp(0.910)=2.485,  exp(0.634)=1.885
sum = 7.704
P(POSITIVE|x) = 3.334/7.704 = 0.433
P(NEUTRAL |x) = 2.485/7.704 = 0.322   <- WAIT, re-check
```

Actually: P(NEUTRAL) wins when z_neu is highest.

```
Re-run: z_neu=0.910 not highest here (z_pos=1.204 wins)
Predicted: POSITIVE  (science discovery article)
```

CONSISTENCY CHECK

```
Article label:     POSITIVE  (exciting discovery)
TF-IDF summary:    POSITIVE  -> MATCH    score +1
BiLSTM summary:    POSITIVE  -> MATCH    score +1
Transformer sum.:  POSITIVE  -> MATCH    score +1
```

WHY EXTRINSIC?

No reference summary needed at all.

Tests whether summary is USEFUL for downstream NLP tasks, not whether it literally matches a specific reference text.

A semantically correct but differently-worded summary

```
will PASS extrinsic eval but MAY FAIL intrinsic ROUGE.
```

RESULTS

```
TF-IDF match rate: ~78%   BiLSTM: ~83%   Transformer: ~87%
```

(4) TOPIC CLASSIFICATION [EXTRINSIC]

NO REFERENCE NEEDED

WHAT IT MEASURES

Does the summary preserve the topic of the article?

Extrinsic: run 10-class topic classifier on article and summary.

CLASSIFIER: RANDOM FOREST (30 trees)

```
Input: TF-IDF vector (~1000 top features)
Classes: Politics Sports Science Technology Business
         Health Entertainment World Arts Environment
Feature subset per split: sqrt(1000) ~ 32 features
Bootstrap sampling: each tree trained on ~63% of data
Final: majority vote across 30 trees
```

DECISION TREE — ONE NODE (entropy-based split)

Parent node S: {Science:4, Tech:3, Politics:3} n=10

```
H(S) = -(4/10*log2(4/10) + 3/10*log2(3/10) + 3/10*log2(3/10))
     = -(0.4*(-1.322) + 0.3*(-1.737) + 0.3*(-1.737))
     = 1.571 bits  (entropy of parent)
```

Candidate split: word "telescope" > 0.3 (TF-IDF threshold)

```
Left  (telescope>0.3): {Sci:3, Tech:1}  n=4
Right (telescope<=0.3): {Tech:2, Pol:3} n=6
H(Left)  = -(3/4*log2(3/4) + 1/4*log2(1/4)) = 0.811 bits
H(Right) = -(2/6*log2(2/6) + 4/6*log2(4/6)) [approx] ~ 0.918 bits
Wait: Right = {Tech:2,Pol:3} n=5 not 6 -> re-check
H(Right) = -(2/5*log2(2/5) + 3/5*log2(3/5)) = 0.971 bits
IG = 1.571 - (4/10*0.811 + 6/10*0.971) = 0.375 bits
Split on "telescope" chosen if IG is best among candidates
```

RANDOM FOREST VOTE

```
Tree 01: "telescope"=0.71, "black hole"=0.68  -> Science
Tree 02: "MIT"=0.82, "discovered"=0.54       -> Science
Tree 03: "billion"=0.61, "mass"=0.58          -> Science
Tree 04: "team"=0.71, "led"=0.49              -> Science
...  (30 trees total)
Majority vote: Science (24/30 trees)  -> Final: SCIENCE
```

CONSISTENCY CHECK

```
Article topic:    SCIENCE   (black hole discovery)
Summary topic:    SCIENCE   -> MATCH  score +1
```

CROSS-VALIDATION

```
5-fold CV accuracy: 86.2% +/- 1.8%
Bootstrap + random features -> prevents overfitting
```

RESULTS

```
TF-IDF: ~81%   BiLSTM: ~85%   Transformer: ~89%
```

INTRINSIC vs EXTRINSIC EVALUATION — KEY DISTINCTION

INTRINSIC (Modules 1 + 2)

Directly compares summary vs reference or article text.

ROUGE: n-gram overlap with human reference summary.

NER: entity recall from original article.

Weakness: valid paraphrases get ZERO credit.

"found" vs "discovered" -> zero ROUGE despite same meaning!

Needs reference summary. Measures surface lexical fidelity.

EXTRINSIC (Modules 3 + 4)

Tests whether summary is USEFUL for a downstream NLP task.

Sentiment: does summary preserve positive/negative tone?

Topic: does summary preserve subject category?

`Strength: no reference summary needed at all.`

`Semantically correct -> passes extrinsic even with 0% ROUGE!`

Measures real-world utility, not surface text similarity.

STAGE 5

METRICS SCORECARD — PER-MODEL COMPARISON

TF-IDF EXTRACTIVE

ROUGE-1 F1:	17.0%
ROUGE-2 F1:	6.1%
ROUGE-L F1:	9.2%
NER Precision:	100%
NER Recall:	~60%
Sentiment Match:	78%
Topic Match:	81%

BiLSTM + BAHDANAU

ROUGE-1 F1:	~20%
ROUGE-2 F1:	~8%
ROUGE-L F1:	~14%
NER Precision:	~90%
NER Recall:	~65%
Sentiment Match:	83%
Topic Match:	85%

TRANSFORMER (BEST)

ROUGE-1 F1:	24.5%
ROUGE-2 F1:	9.8%
ROUGE-L F1:	19.9%
NER Precision:	~90%
NER Recall:	0%*
Sentiment Match:	87%
Topic Match:	89%

* Transformer NER Recall=0%: BPE vocabulary lowercased -> "earth" != "Earth" -> spaCy misses entity. Fix: capitalise tokens or use neural NER.

STREAMLIT DASHBOARD — INTERACTIVE EVALUATION INTERFACE

N-GRAM PANEL

- Bar chart: ROUGE-1/2/L
- for all 3 models side-by-side
- Relative gain labels
- "+117% on ROUGE-L" annotation
- TF-IDF vs Transformer
- Download results as CSV

NER PANEL

- Entity table per model
- PERSON ORG GPE FACILITY types
- Highlight: missed entities
- Precision/Recall bar chart
- Casing bug warning flag
- Show article entities list

SENTIMENT PANEL

- Pie chart: POS/NEU/NEG
- Article vs Summary comparison
- Match indicator check/cross
- Softmax probability bars
- z-score display per class
- LR decision boundary plot

TOPIC PANEL

- Predicted topic chip
- Confidence per class bar
- Decision tree visualized
- 30-tree vote histogram
- 5-fold CV accuracy: 86.2%
- Feature importance ranking